Bo Pang

Email: pangbo_zju@zju.edu.cn — Phone: +86-13176666917

Education

Zhejiang University, Hangzhou, China — Sep 2023 – Present
Ph.D. in Control Science and Engineering
Advisors: Prof. Liang Li and Prof. Jiming Chen

Shandong University, Jinan, China — Sep 2019 – Jun 2023
B.Eng. in Automation
Advisors: Prof. Chaoqun Wang, Prof. Yan Li, and Prof. Bo Sun
GPA: 96.69 / 100

Research Interests

Vision-Language Navigation; Spatial-Semantic Grounding; Map-light Autonomous Navigation; LiDAR-based Place Recognition and Sparse-map Retrieval; 3D Perception.

Publications

Published Papers

- **Bo Pang**, Yang Xu, Jiming Chen, and Liang Li. "PB-MOT: Pose-aware Association Boosted Online 3D Multi-Object Tracking." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

Manuscripts

- **Bo Pang**, Yang Xu, Jiming Chen, and Liang Li. "PB-MOT++: Extending Pose-aware Association to Trajectory-Centric Offline Optimization." *Manuscript under review.*
- **Bo Pang**, et al. "Robust Place Recognition on Sparse Maps via a Semantic-Enhanced Neural Network." *Manuscript in revision for journal resubmission.*

Research Experience

Vision-Language Navigation with Coarse Maps and Semantic Landmarks — 2026 – Present
Zhejiang University
- Studying how navigation signs, object landmarks, and scene-level semantic structures can serve as reliable anchors for language-guided localization and route reasoning.
- Exploring spatial-semantic representations that connect geometric perception, place recognition, and language-conditioned navigation decisions in map-light robotic systems.
- Extending previous research experience in LiDAR-based place recognition and loop closure detection toward more interactive, instruction-aware, and cognitively grounded autonomous navigation.

Sparse-map Place Recognition and Loop Closure Detection — 2025 – 2026
Zhejiang University
- Investigated LiDAR-based place recognition on sparse maps, treating sparse maps as compact and deployment-oriented representations for map-light navigation.

- Developed semantic-enhanced bird's-eye-view representations that integrate geometric occupancy and semantic cues for robust sparse-map retrieval.
- Explored rotation- and translation-tolerant global descriptors for large-displacement loop closure detection under sparse and incomplete map observations.

Trajectory-Centric 3D Multi-Object Tracking 2023 – 2025
Zhejiang University

- Developed pose-aware association strategies for online 3D multi-object tracking, improving identity consistency through ego-motion-aware spatial reasoning.
- Achieved leaderboard-leading performance on the KITTI 3D Multi-Object Tracking benchmark in both online and offline settings, **ranking first among submitted methods** at the time of evaluation.
- Extended online tracking into a trajectory-centric offline optimization framework for evidence recovery, topology repair, gap completion, and kinematic state refinement.

Awards and Honors

- **Second Prize**, 16th China Undergraduate Intelligent Vehicle Competition 2021
  Basic Four-Wheel Category, National Finals
- **Second Prize**, 15th China Undergraduate Intelligent Vehicle Competition 2020
  Dual-Car Relay Category, National Finals
- **Second Prize**, National Electronic Design Contest for College Students, Shandong Division 2020, 2021

Technical Skills

- **Programming:** Python, C++, MATLAB
- **Robotics and Perception:** ROS, OpenCV, OpenPCDet, SLAM, 3D Object Detection, Multi-Object Tracking
- **Tools:** Git, Linux, LaTeX

Academic Service

- Teaching Assistant, *Robotics II: Mobile Robotics*, Department of Control Science and Engineering, Zhejiang University Sep 2024 – Jan 2025
- Teaching Assistant, *Intelligent Mobile Systems*, Department of Control Science and Engineering, Zhejiang University Sep 2023 – Jan 2024
- Teaching Assistant, SDG-NAS Summer School on Networked Autonomous Systems, Zhejiang University Jul 2025
- Teaching Assistant, SDG-NAS Summer School on Networked Autonomous Systems, Zhejiang University Aug 2024
- Volunteer, National PhD Student Academic Forum in Control Science and Engineering, Zhejiang University Nov 2024